MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
FOURTH SEMESTER EXAMINATION
(2024 ADMISION ONWARDS)
MG4DSEMAT202: OPERATIONS RESEARCH

Duration: 2 hrs Maximum Marks: 70

*Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.*

**Part A**
**Answer any five questions. Each question carries 2 marks**

1. Give the canonical form of a Linear Programming problem. [K][1]
2. What is mean by infeasible solution? [K][1]
3. Define degenerate basic feasible solution. [K][3]
4. Give any two rules to construct the dual of a LPP. [K][4]
5. Give any two assumptions in the transportation model. [K][5]
6. What is mean by degeneracy in transportation problem? [U][5]
7. Give the mathematical representation of the assignment model [U][6]
8. What is mean by unbalanced assignment problem? [U][6]

**Part B**
**Answer any five questions. Each question carries 6 marks**

9. A company has two grades of inspectors, I and II to undertake quality contril inspection. At least 1500 pieces must be inspected in an 8- hour day. Grade I inspector can check 20 pieces in an hour with an accuracy of 96%. Grade II inspector checks 14 pieces an hour with an accuracy of 92%. Wages of grade I inspector are Rs. 5 per hour while those of grade II inspector are Rs.4 per hour. Any error made by an inspector costs Rs.3 to the company. If there are in all 10 grade I inspectors and 15 grade II inspectors in the company, formulate a LPP to find the optimal assignment of inspectors that minimizes the daily inspection cost.[U,A][1]
10. Find the mimimum value of $Z = 5x_1 - 2x_2$
subject to $2x_1 + 3x_2 \geq 1, x_1, x_2 \geq 0$. [U, A][2]

11. Find the initial basic feasible solution of the following LPP.
$MinimizeZ = x_1 - 3x_2 + 3x_3$
subject to $3x_1 - x_2 + 2x_3 \leq 7$,
$2x_1 + 4x_2 \geq -12$,
$-4x_1 + 3x_2 + 8x_3 \leq 10$,
$x_1, x_2, x_3 \geq 0$. [A][3]

12. Construct the dual of the following LPP.
Maximize $Z = 3x_1 + 10x_2 + 2x_3$
Subject to $2x_1 + 3x_2 + 2x_3 \leq 7$
$3x_1 - 2x_2 + 4x_3 = 3$
$x_1, x_2, x_3 \geq 0$ [A][4]

13. Find the initial basic feasible solution using MODI method.

| origin/destination | 1 | 2 | 3 | 4 | 5 | 6 | Supply |
|---|---|---|---|---|---|---|---|
| 1 | 9 | 12 | 9 | 6 | 9 | 10 | 5 |
| 2 | 7 | 3 | 7 | 7 | 5 | 5 | 6 |
| 3 | 6 | 5 | 9 | 11 | 3 | 11 | 2 |
| 4 | 6 | 8 | 11 | 2 | 2 | 10 | 9 |
| Demand | 4 | 4 | 6 | 2 | 4 | 2 | |

[A][5]

14. Give the transportation algorithm for a minimization problem [K][5]

15. Compare assignment model with transportation model [U][5,6]

16. Define the assignment model and give the mathematical representation of the assignment model. [K,U][6]

**Part C**
**Answer any three questions.**
**Each question carries 10 marks**

17. Solve graphically. Max $Z = 3x + 2y$
subject to $-2x + 3y \leq 9$
$3x - 2y \leq -20$
$x, y \geq 0$ [A][2]

18. Use simplex method to solve the following LPP
Max $Z = 2x_1 + 5x_2$
Subject to $x_1 + 4x_2 \leq 24$,
$3x_1 + x_2 \leq 21$
$x_1 + x_2 \leq 9$,
$x_1, x_2 \geq 0$. [A][3]

19. Use the two-phase simplex method to
Max $Z = 5x_1 + 3x_2$
subject to $2x_1 + x_2 \leq 1$,
$x_1 + 4x_2 \geq 6$
$x_1, x_2 \geq 0$ [A][3]

20. Find the solution of the following transportation problem.

| origin/destination | 1 | 2 | 3 | 4 | Supply |
|---|---|---|---|---|---|
| A | 4 | 6 | 8 | 13 | 50 |
| B | 13 | 11 | 10 | 8 | 70 |
| C | 14 | 4 | 10 | 13 | 30 |
| D | 9 | 11 | 13 | 8 | 50 |
| Demand | 25 | 35 | 105 | 20 | |

[A][5].

21. Find the optimum solution of the following transportation poblem

| origin/destination | W1 | W2 | W3 | W4 | W5 | Available |
|---|---|---|---|---|---|---|
| F1 | 7 | 6 | 4 | 5 | 9 | 40 |
| F2 | 8 | 5 | 6 | 7 | 8 | 30 |
| F3 | 6 | 8 | 9 | 6 | 5 | 20 |
| F4 | 5 | 7 | 7 | 8 | 6 | 10 |
| Required | 30 | 30 | 15 | 20 | 5 | |

[A][5]

22. Solve the following assignment problem

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 2 | 0 | 1 | 4 |
| 2 | 0 | 1 | 4 | 2 |
| 3 | 3 | 2 | 0 | 1 |
| 4 | 1 | 3 | 0 | 2 |

[A][6]